LAPORAN PRAKTIKUM
TEKNOLOGI IOT II

MODUL 6: PROTOKOL KOMUNIKASI TELEMETRI INDUSTRI (MQTT)

Disusun Oleh :

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Kelas : TRI - 3D

Dosen Pengampu :

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Agus Ambarwari, S.Pd., M.Kom.

TEKNOLOGI REKAYASA INTERNET

JURUSAN TEKNOLOGI INFORMASI

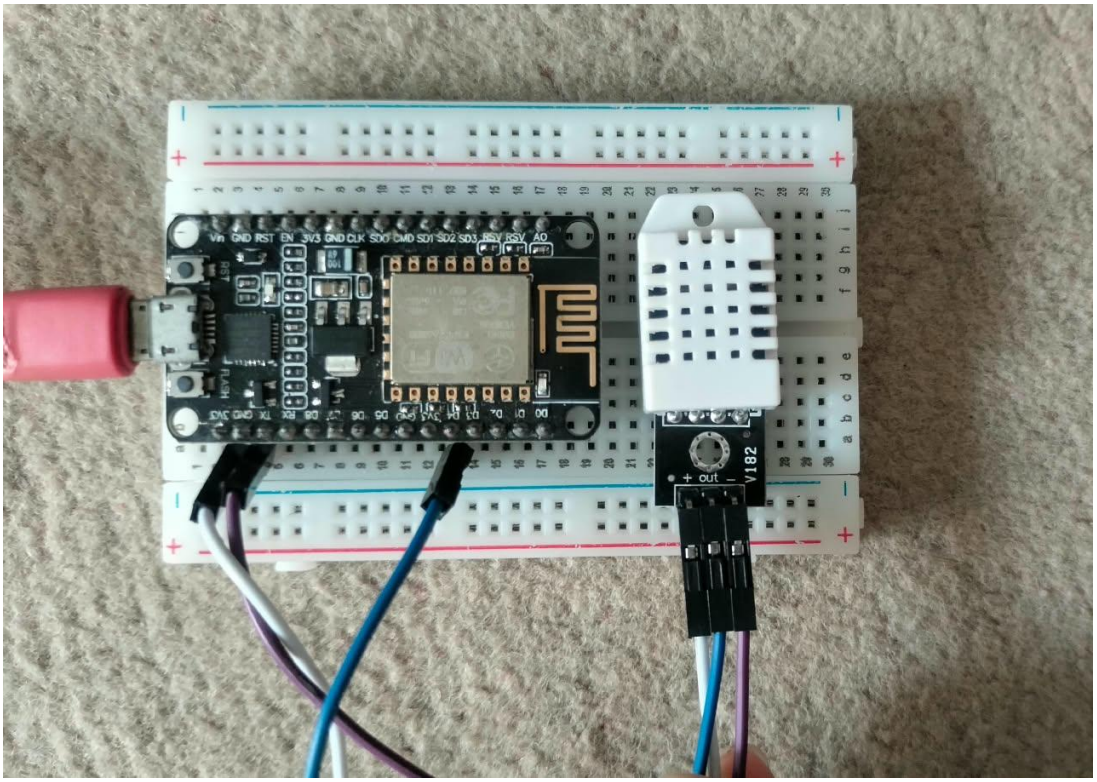
POLITEKNIK NEGERI LAMPUNG

2026/2027

6.5 Praktikum

6.5.1 Praktikum 1: Publikasi Data Sensor Lingkungan (DHT) Berformat JSON

Rangkaian



Kode

6.5.1_Praktikum_1.ino

```
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include <DHT.h>

// Kredensial Jaringan Wi-Fi (Gunakan Hotspot Pribadi Anda)
const char* ssid = "kar";
const char* password = "12345678";

// Konfigurasi Broker MQTT Publik
const char* mqttServer = "broker.hivemq.com";
const int mqttPort = 1883;

// IDENTITAS KLIEN (Wajib Unik! Ganti dengan NIM Anda agar tidak bertabrakan)
const char* clientId = "ESP8266_LabIoT_25782112";

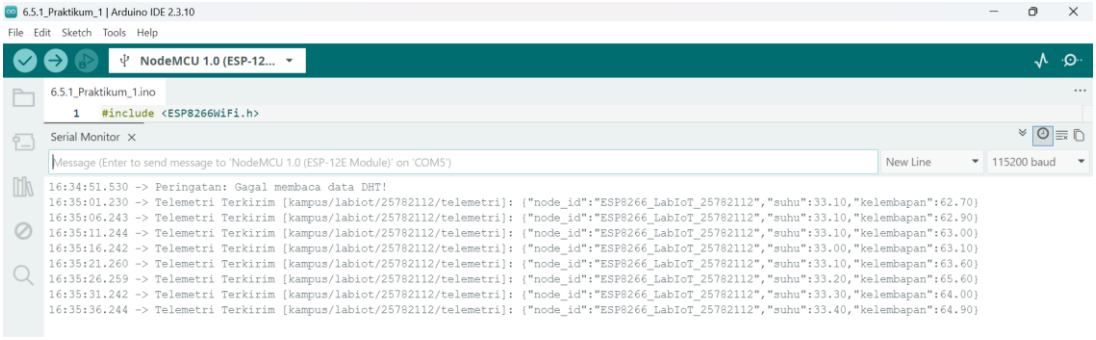
// Definisi Topik Publikasi Telemetri Mandiri
const char* topicPubTelemetry = "kampus/labiot/25782112/telemetri";

// Konfigurasi Pin Sensor DHT
const byte dhtPin = 2;   // Pin D4 (GPIO 2)
#define DHTTYPE DHT22

DHT dht(dhtPin, DHTTYPE);
WiFiClient espClient;
PubSubClient client(espClient);

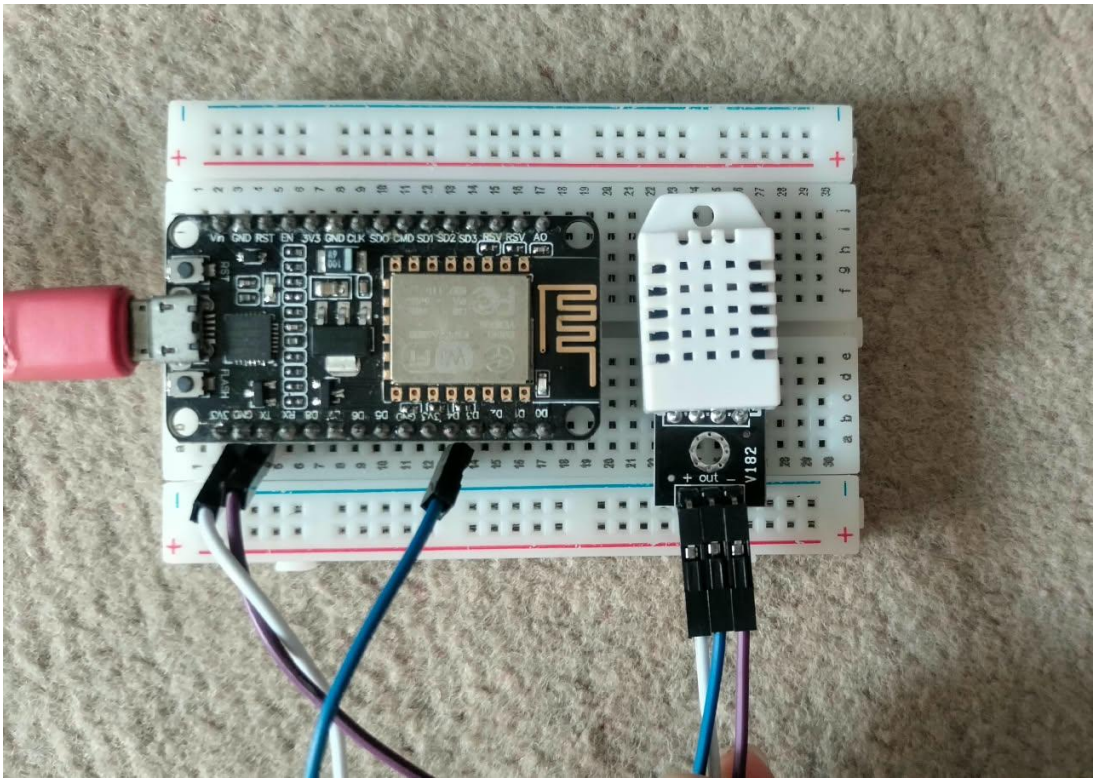
unsigned long previousPublishTime = 0;
const unsigned long publishInterval = 5000; // Interval pengiriman 5 detik
```

Serial monitor



6.5.2 Praktikum 2: Subskripsi dan Kontrol Indikator LED Asinkron

Rangkaian



Kode

6.5.2_Praktikum_2.ino

```
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>

// Kredensial Jaringan Wi-Fi
const char* ssid = "kar";
const char* password = "12345678";

// Konfigurasi Broker MQTT Publik
const char* mqttServer = "broker.hivemq.com";
const int mqttPort = 1883;
const char* clientId = "ESP8266_LabIoT_25782112_Sub";

// Definisi Topik Subskripsi Perintah Mandiri
const char* topicSubCommand = "kampus/labiot/25782112/perintah";

// Konfigurasi Pin Indikator LED
const byte ledPin = 5; // Pin D1 (GPIO 5)

WiFiClient espClient;
PubSubClient client(espClient);

void connectToWiFi() {
  Serial.print("Menghubungkan ke Wi-Fi: ");
  Serial.println(ssid);
  WiFi.mode(WIFI_STA);
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
```

Serial Monitor

6.5.2_Praktikum_2.ino

```
#include <ESP8266WiFi.h>
```

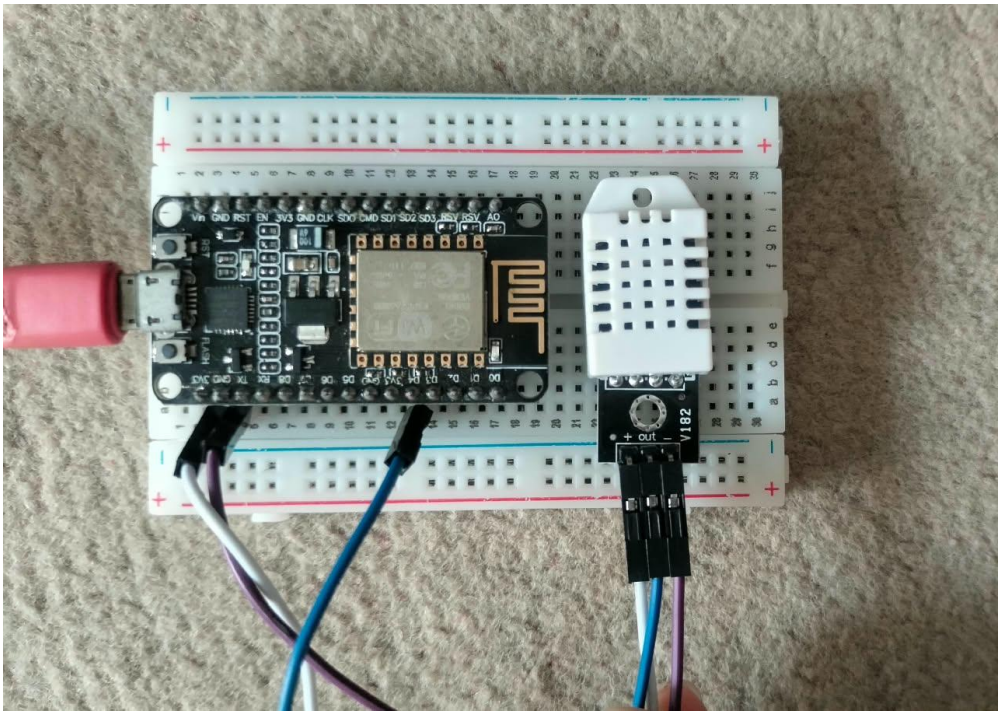
Output Serial Monitor ×

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')

```
08:25:50.124 -> rl□l□□|□□l□|□□□□$□b|□□□□□□|□b□b□□oN□□$no□□□□b
08:25:58.518 -> ..................
08:26:07.773 -> Wi-Fi Terkoneksi!
08:26:07.773 -> Mencoba koneksi MQTT... Terhubung!
08:26:09.125 -> Subskripsi Topik: kampus/labiot/25782112/perintah
```

6.5.3 Praktikum 3: Sistem Terpadu Dua Arah (Full Duplex Telemetry & Control)

Rangkaian



Kode

6.5.3_Praktikum_3.ino

```
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include <DHT.h>

// Kredensial Jaringan Wi-Fi
const char* ssid = "kar";
const char* password = "12345678";

// Konfigurasi Broker MQTT Publik
const char* mqttServer = "broker.hivemq.com";
const int mqttPort = 1883;
const char* clientId = "ESP8266_LabIoT_25782112_Full";

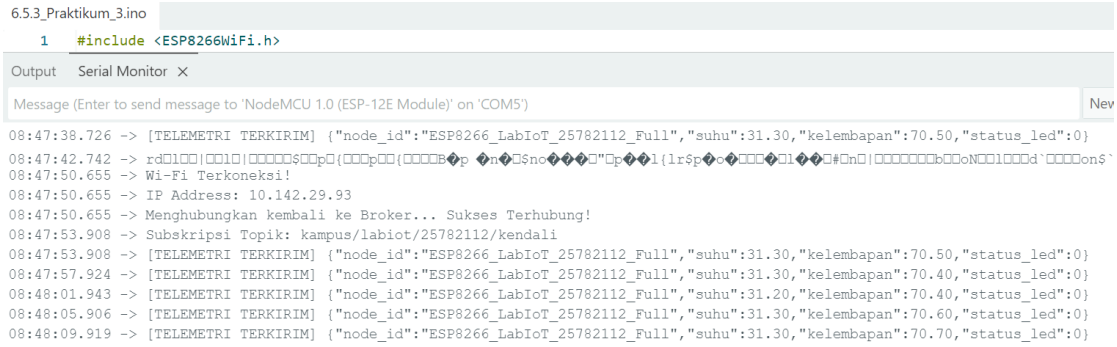
// Pembagian Topik Mandiri
const char* topicPublish   = "kampus/labiot/25782112/telemetri";
const char* topicSubscribe = "kampus/labiot/25782112/kendali";

// Alokasi Pin Perangkat Keras
const byte dhtPin = 2; // Pin D4 (GPIO 2)
const byte ledPin = 5; // Pin D1 (GPIO 5)
#define DHTTYPE DHT22

DHT dht(dhtPin, DHTTYPE);
WiFiClient espClient;
PubSubClient client(espClient);

unsigned long lastTelemetryTime = 0;
const unsigned long telemetryInterval = 4000; // Siklus kirim 4 detik
```

Serial Monitor

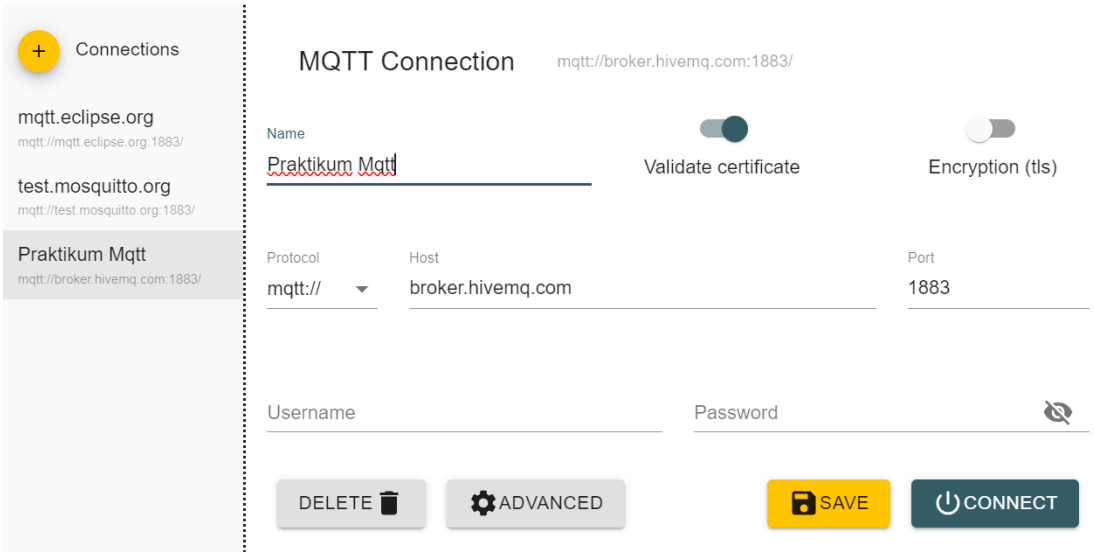


6.5.4 Panduan Pengujian Diagnostik dengan MQTT Explorer

1. Jalankan Aplikasi MQTT Explorer.

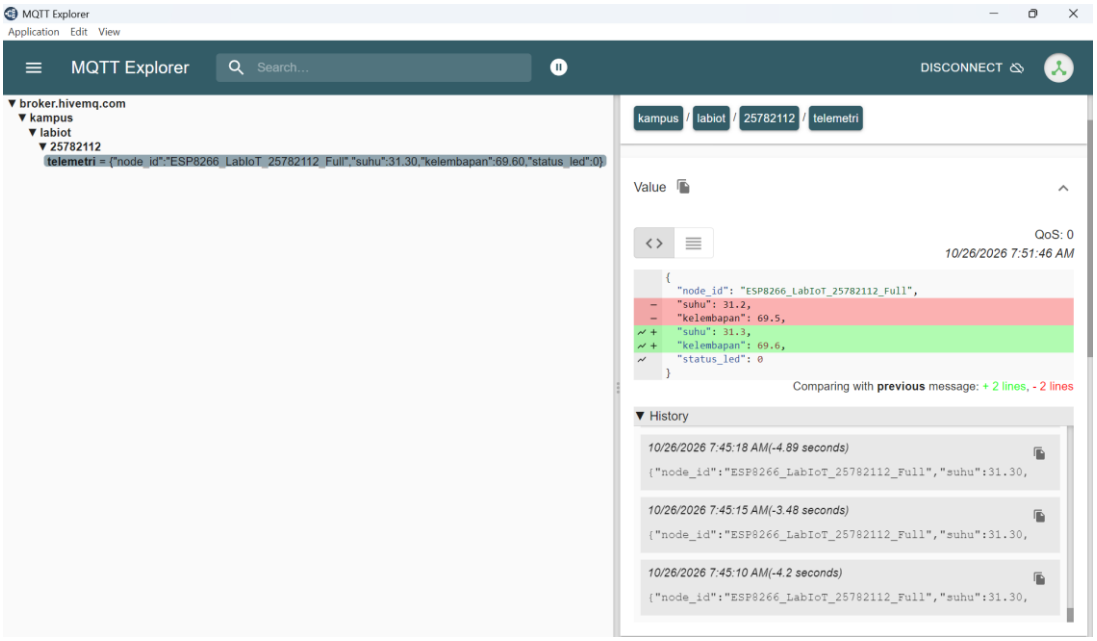
2. Lalu isi konfigurasi koneksi Broker seperti berikut:

- Protocol: mqtt://
- Host: broker.hivemq.com
- Port: 1883
- Klik tombol Connect.



3. Memantau Data Masuk (Telemetry Monitoring):

Amati pohon hierarki topik (Topic Tree) pada panel sebelah kiri. Temukan dan klik cabang topik Anda: kampus → labiot → nim_anda → telemetri. Anda akan melihat payload JSON diperbarui secara real-time setiap 4 detik. Klik pada nilai numerik untuk menampilkan grafik fluktuasi suhu dan kelembapan secara langsung.

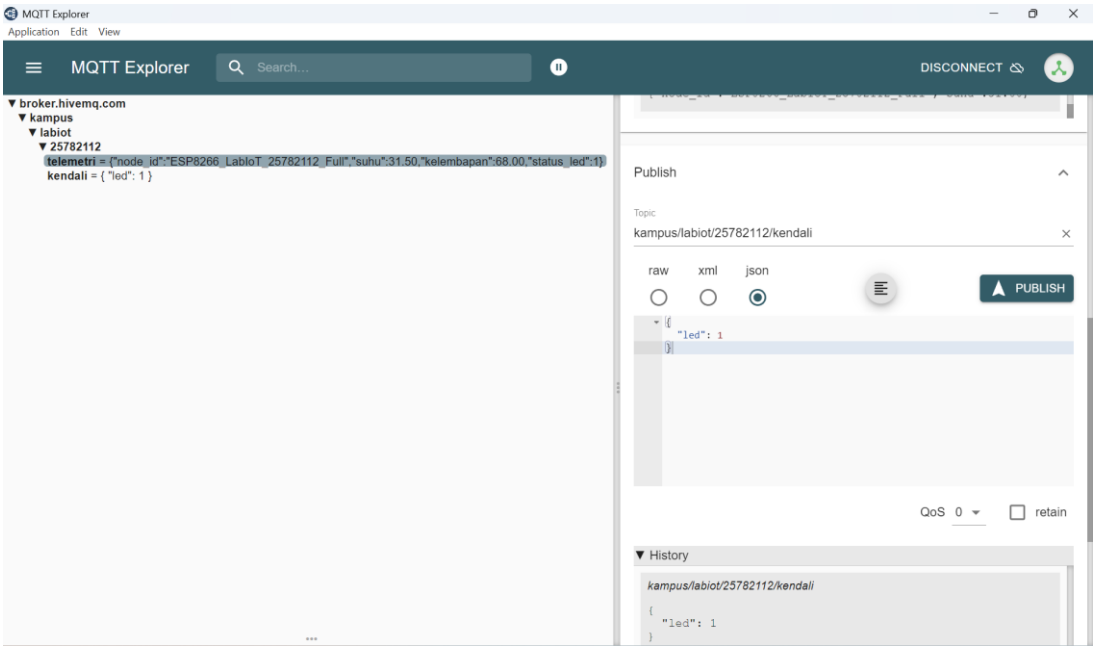


4. Menerbitkan Instruksi Kendali (*Publish Command*):

- Arahkan ke panel Publish di sisi kanan bawah antarmuka.
- Masukkan topik tujuan: kampus/labiot/nim_anda/kendali
- Pilih format muatan: JSON.
- Masukkan muatan payload berikut:
- {

  "led": 1

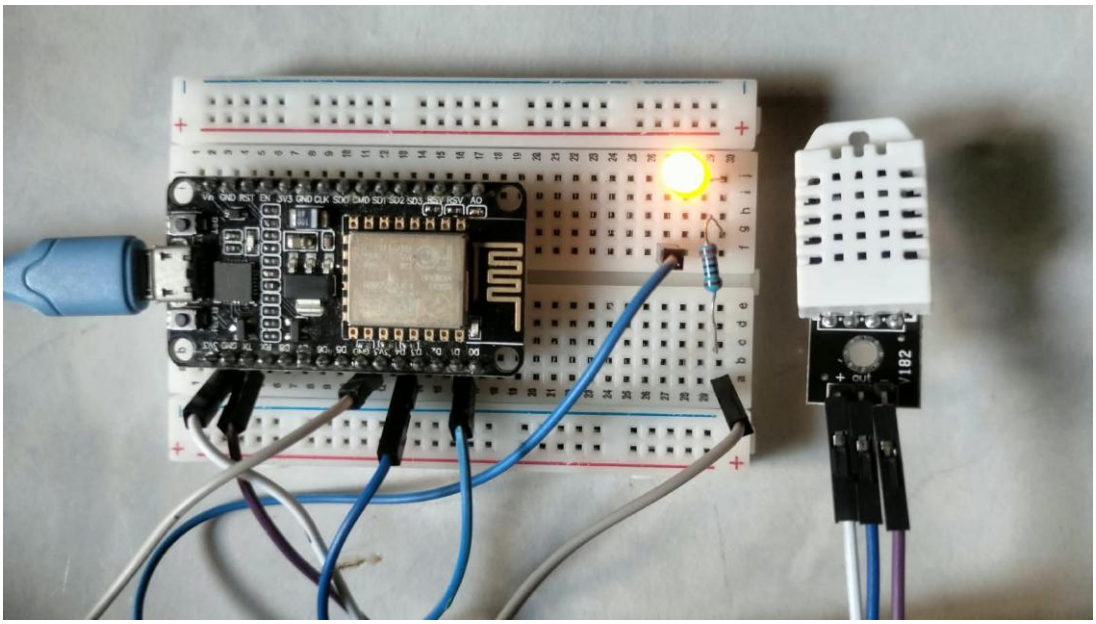
  }
- Klik tombol Publish. Perhatikan bahwa LED indikator menyala seketika tanpa menghentikan siklus pengiriman data telemetri.



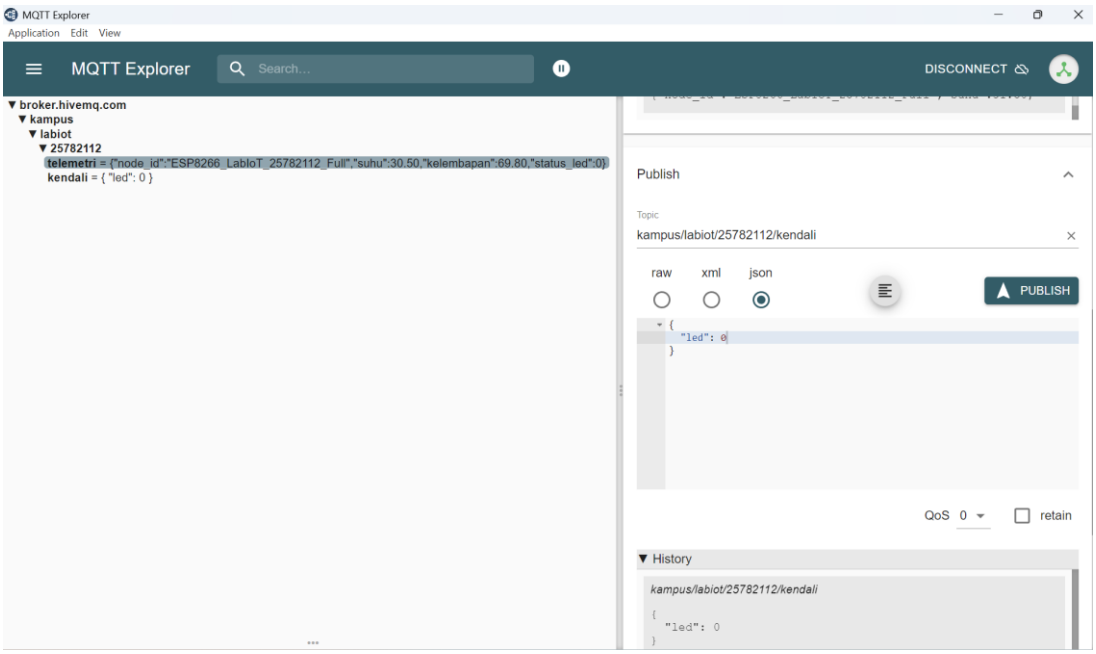
Serial Monitor

```
07:54:46.503 -> ----------------------------------------
07:54:46.503 -> [KONTROL DITERIMA] Topik: kampus/labiot/25782112/kendali | Payload: {
07:54:46.503 ->   "led": 1
07:54:46.503 -> }
07:54:46.503 -> --> [INDIKATOR] Status LED diubah: MENYALA
07:54:46.503 -> ----------------------------------------
07:54:50.286 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":68.30,"status_led":1}
07:54:54.252 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":68.10,"status_led":1}
07:54:58.252 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":68.20,"status_led":1}
07:55:02.270 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":67.80,"status_led":1}
07:55:06.255 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":68.00,"status_led":1}
07:55:10.251 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":67.80,"status_led":1}
07:55:14.271 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":67.70,"status_led":1}
07:55:18.294 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":67.80,"status_led":1}
07:55:22.260 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":67.80,"status_led":1}
07:55:26.287 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":67.80,"status_led":1}
07:55:30.253 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":67.80,"status_led":1}
07:55:34.271 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":67.80,"status_led":1}
```

Kondisi Led hidup pada rangkaian



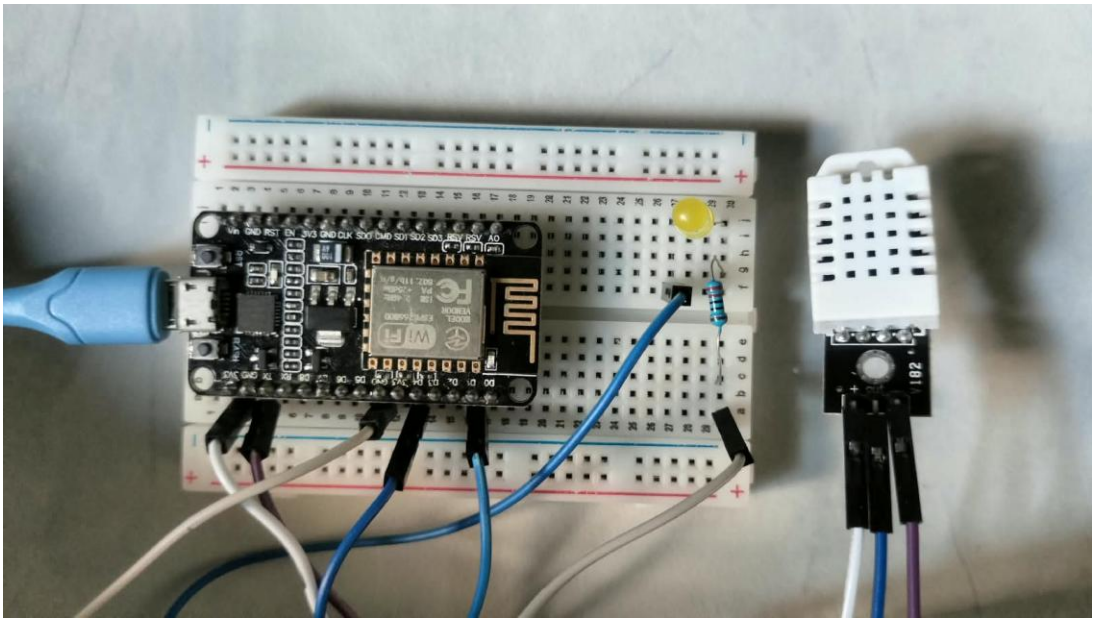
- Kirimkan perintah pematian dengan payload {"led": 0} dan pastikan LED padam.



Serial Monitor

```
08:00:41.289 -> ----------------------------------------
08:00:41.289 -> [KONTROL DITERIMA] Topik: kampus/labiot/25782112/kendali | Payload: {
08:00:41.323 ->   "led": 0
08:00:41.323 -> }
08:00:41.323 -> --> [INDIKATOR] Status LED diubah: PADAM
08:00:41.323 -> ----------------------------------------
08:00:42.295 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.50,"kelembapan":69.80,"status_led":0}
08:00:46.267 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.50,"kelembapan":69.80,"status_led":0}
08:00:50.259 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.50,"kelembapan":69.80,"status_led":0}
08:00:54.265 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.50,"kelembapan":69.70,"status_led":0}
08:00:58.295 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.50,"kelembapan":69.70,"status_led":0}
08:01:02.267 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.50,"kelembapan":69.70,"status_led":0}
08:01:06.291 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.50,"kelembapan":69.80,"status_led":0}
```

Kondisi Led padam pada rangkaian



6.6 Latihan

Kerjakan latihan berikut secara mandiri berbasis analisis teori dan modifikasi kode program untuk melengkapi laporan praktikum Anda:

- **Latihan 1 (Analisis Arsitektur & Perbandingan Protokol):**
  Pada Modul 3, kita menggunakan protokol HTTP untuk membaca dan mengontrol perangkat. Jelaskan secara mendalam mengapa protokol MQTT jauh lebih unggul daripada HTTP dalam hal:
  i. Efisiensi penggunaan kapasitas *bandwidth* pada paket data seluler.
  ii. Beban komputasi mikrokontroler (*CPU overhead*) saat menangani pertukaran data dua arah.
  iii. Kemampuan pengiriman data secara simultan ke banyak perangkat penerima (*1-to-N distribution*).

  Jawab:

  MQTT lebih unggul dibandingkan HTTP untuk sistem IoT karena lebih ringan dan efisien. Dari sisi bandwidth, MQTT memiliki overhead paket yang lebih kecil sehingga lebih hemat saat mengirim data sensor secara berkala. Dari sisi beban mikrokontroler, MQTT menggunakan komunikasi asinkron melalui broker dan callback sehingga ESP8266 tidak perlu terus melakukan request seperti pada HTTP. Selain itu, MQTT mendukung distribusi 1-to-N, yaitu satu publisher cukup mengirim data ke broker, lalu broker meneruskannya ke banyak subscriber yang berlangganan topic yang sama. Dengan cara ini, komunikasi menjadi lebih sederhana, ringan, dan cocok untuk sistem IoT dengan banyak perangkat.
- **Latihan 2 (Analisis Hierarki Topik & Karakter Wildcard):**
  Sebuah pabrik manufaktur memiliki 3 lini produksi (*line1*, *line2*, *line3*), di mana setiap lini memiliki 4 mesin (*m1*, *m2*, *m3*, *m4*), dan setiap mesin melaporkan metrik suhu, getaran, dan status. Struktur topiknya adalah pabrik/<line>/<mesin>/<sensor>. Tentukan ekspresi topik subskripsi (*subscription topic*) yang tepat menggunakan karakter *wildcard* (+ atau #) untuk:

i. Memantau parameter suhu dari seluruh mesin di semua lini produksi.
ii. Memantau seluruh parameter sensor (suhu, getaran, status) khusus untuk mesin m1 di lini line2.
iii. Memantau seluruh lalu lintas data tanpa terkecuali dari seluruh lini dan seluruh mesin di pabrik tersebut.

Jawab:

Pada MQTT, wildcard + digunakan untuk menggantikan satu level topic, sedangkan # digunakan untuk mencakup seluruh level setelahnya.

1. Untuk memantau suhu dari seluruh mesin di semua lini:
   pabrik/+/+/suhu
2. Untuk memantau seluruh parameter sensor pada mesin m1 di line2:
   pabrik/line2/m1/#
3. Untuk memantau seluruh data dari semua lini dan mesin:
   pabrik/#

   Dengan wildcard tersebut, proses monitoring menjadi lebih sederhana karena tidak perlu melakukan subscribe ke setiap topic satu per satu.

- **Latihan 3 (Implementasi Serialisasi Data Dinamis):**

  Modifikasi blok penyusunan JSON pada Praktikum 1 agar paket telemetri menyertakan dua parameter tambahan berikut:

  i. "uptime_detik": Lama waktu aktif sistem dalam satuan detik (dihitung dari rumus millis() / 1000).
  ii. "ip_lokal": Alamat IP lokal mikrokontroler dalam format string (WiFi.localIP().toString()).

  Tuliskan baris modifikasi kode C++ yang Anda tambahkan pada fungsi pembentukan JSON!

  Jawab:

  Pada Praktikum 1, data telemetri ditambahkan dengan informasi lama waktu aktif perangkat dan alamat IP lokal. Baris kode yang ditambahkan pada proses pembentukan JSON adalah:

  ```
  doc["uptime_detik"] = millis() / 1000;
  doc["ip_lokal"] = WiFi.localIP().toString();
  ```

Dengan penambahan tersebut, payload MQTT tidak hanya berisi suhu dan kelembapan, tetapi juga waktu aktif ESP8266 dalam detik serta alamat IP lokal perangkat.

Kode

```
88    // Alokasi dokumen JSON statis
89    StaticJsonDocument<256> doc;
90    doc["node_id"] = clientId;
91    doc["suhu"] = serialized(String(t, 2));       // Format desimal 2 angka di belakang koma
92    doc["kelembapan"] = serialized(String(h, 2));
93
94    doc["uptime_detik"] = millis() / 1000;
95    doc["ip_lokal"] = WiFi.localIP().toString();
96
```

Serial Monitor



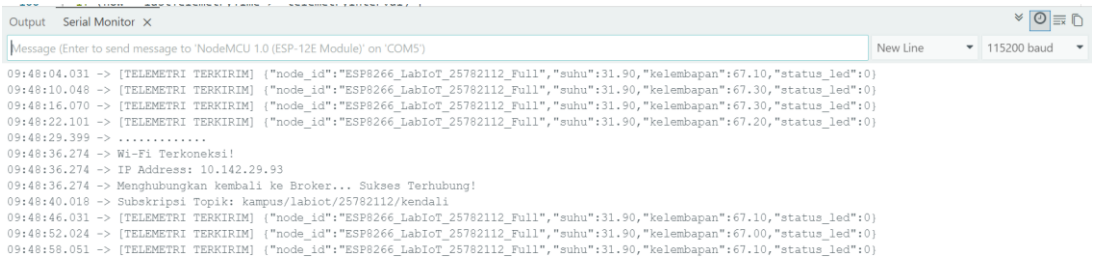
- **Latihan 4 (Analisis Mekanisme Reconnect & Loop Blocking):**
  Pada Praktikum 3, di dalam fungsi loop() terdapat pemanggilan rutin client.loop().
  i. Apa konsekuensi teknis yang terjadi terhadap responsivitas penerimaan perintah *callback* jika seorang pengembang menyisipkan instruksi delay(6000) di dalam fungsi loop()?
  ii. Jelaskan bagaimana fungsi reconnect() bekerja memulihkan koneksi saat router Wi-Fi atau broker MQTT sempat terputus sesaat!

Jawab:

i. Jika delay(6000) ditambahkan di dalam loop(), ESP8266 akan berhenti menjalankan proses selama 6 detik sehingga client.loop() tidak dapat memproses pesan MQTT. Akibatnya, penerimaan perintah melalui callback menjadi terlambat dan koneksi MQTT dapat menjadi kurang responsif.

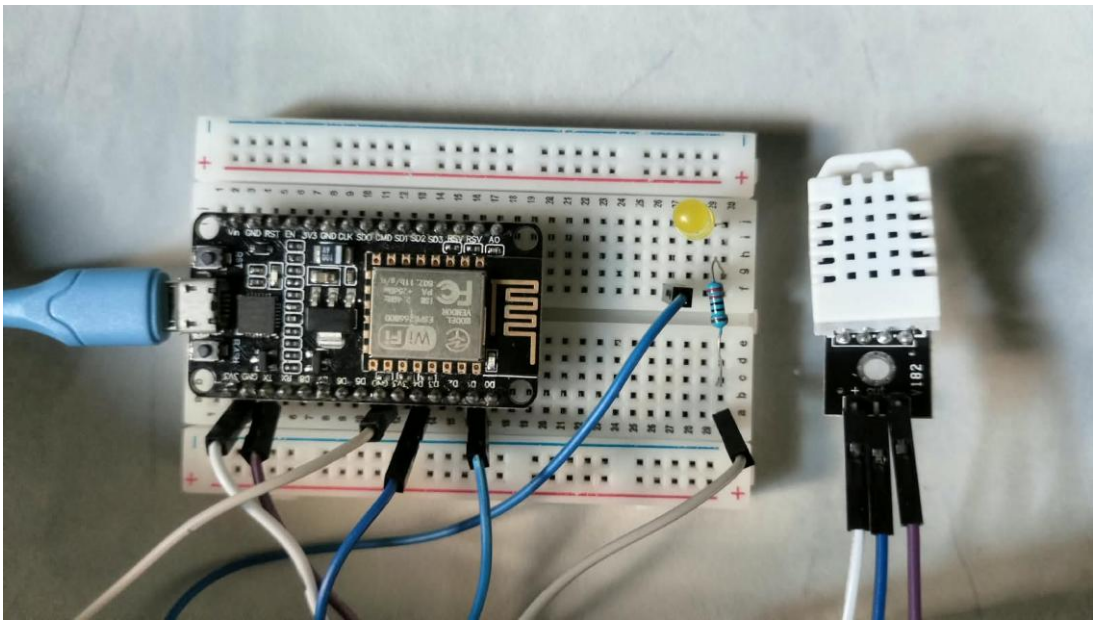
ii. Fungsi reconnect() digunakan untuk memulihkan koneksi saat MQTT terputus. ESP8266 akan mencoba terhubung kembali ke broker menggunakan client.connect(). Setelah berhasil terhubung, perangkat melakukan subscribe ulang ke topic kendali agar dapat kembali menerima pesan dari broker.

Bukti di serial monitor



6.7 Tugas: Sistem Telemetri & Kendali Closed-Loop Industri

Rangkaian



Kode

tugas_prak06.ino

```ino
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include <DHT.h>

// Kredensial Jaringan Wi-Fi
const char* ssid = "kar";
const char* password = "12345678";

// Konfigurasi Broker MQTT Publik
const char* mqttServer = "broker.hivemq.com";
const int mqttPort = 1883;
const char* clientId = "ESP8266_LabIoT_25782112_Full";

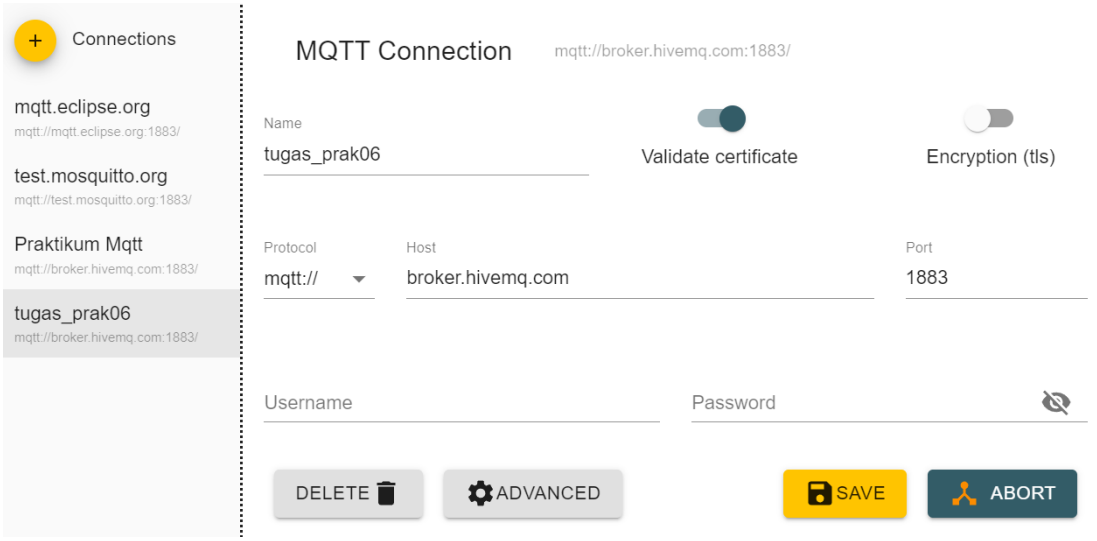
// Pembagian Topik Mandiri
const char* topicPublish   = "kampus/labiot/25782112/telemetri";
const char* topicSubscribe = "kampus/labiot/25782112/kendali";
const char* topicAlert = "kampus/labiot/25782112/alert";
const char* topicStatus = "kampus/labiot/25782112/status";

// LWT
const char* lwtMessage = "{\"status\":\"OFFLINE\",\"keterangan\":\"Perangkat Terputus Tak Terduga\"}";
const char* onlineMessage = "{\"status\":\"ONLINE\",\"keterangan\":\"Perangkat Beroperasi Normal\"}";

// Alokasi Pin Perangkat Keras
const byte dhtPin = 2; // Pin D4 (GPIO 2)
const byte ledPin = 5; // Pin D1 (GPIO 5)
#define DHTTYPE DHT22

DHT dht(dhtPin, DHTTYPE);
WiFiClient espClient;
PubSubClient client(espClient);
```

Koneksi mqtt





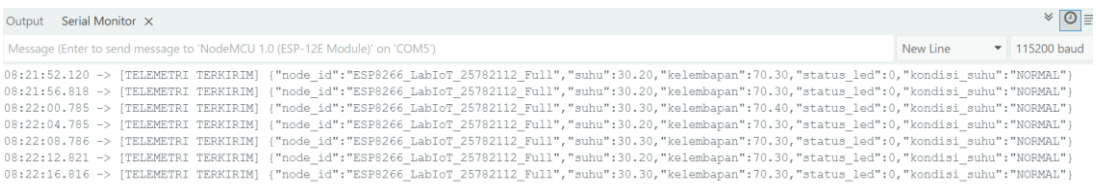
Kriteria Keberhasilan:

1. Data telemetri periodik terkirim stabil setiap siklus waktu dengan field klasifikasi kondisi lingkungan ("NORMAL" / "PANAS").

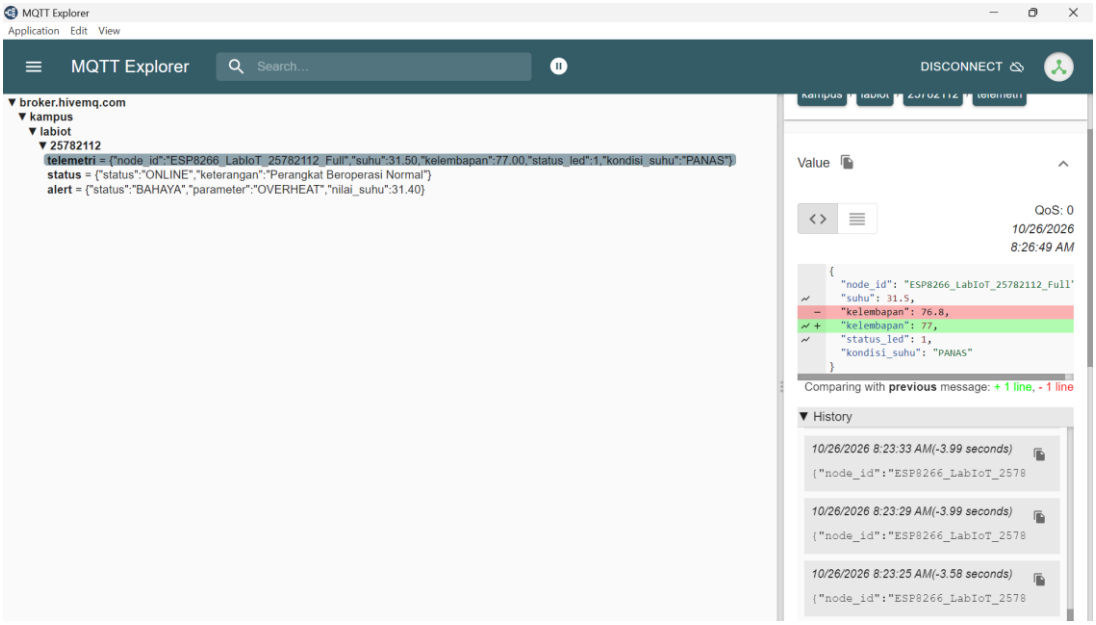
- JIKA suhu ≤ 31.0°C, isi nilai menjadi "NORMAL"



Serial monitor



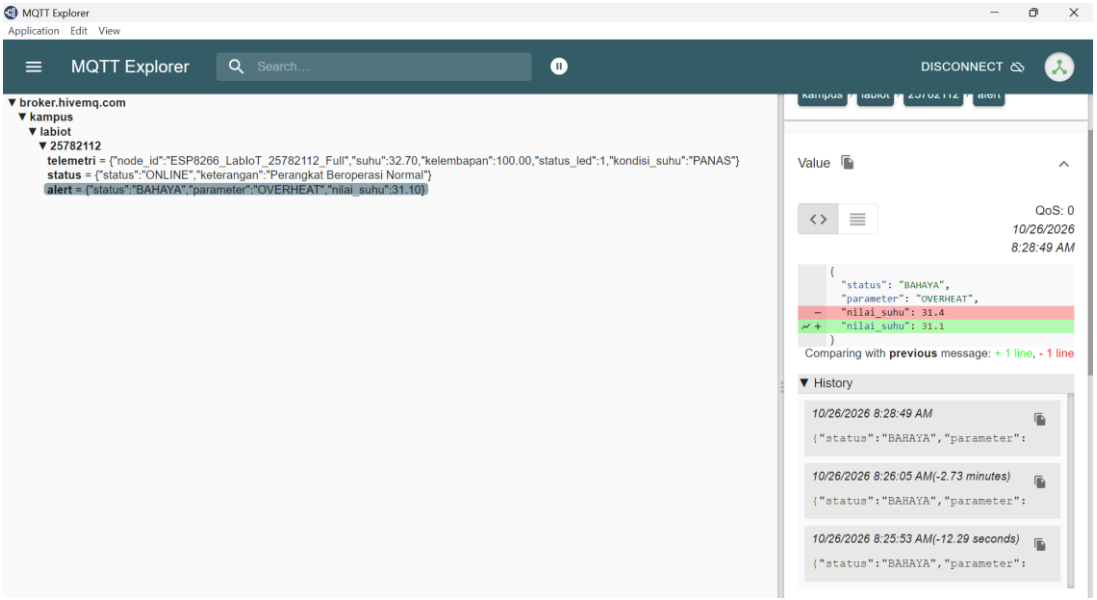
- JIKA suhu > 31.0°C, isi nilai menjadi "PANAS".



Serial Monitor

```
08:25:52.803 -> [ALERT TERKIRIM] {"status":"BAHAYA","parameter":"OVERHEAT","nilai_suhu":31.10}
08:25:52.803 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.10,"kelembapan":98.60,"status_led":1,"kondisi_suhu":"PANAS"}
08:25:56.825 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.70,"kelembapan":97.20,"status_led":1,"kondisi_suhu":"NORMAL"}
08:26:00.825 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.60,"kelembapan":86.80,"status_led":1,"kondisi_suhu":"NORMAL"}
08:26:04.835 -> [ALERT TERKIRIM] {"status":"BAHAYA","parameter":"OVERHEAT","nilai_suhu":31.40}
08:26:04.835 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.40,"kelembapan":94.60,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:08.790 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":94.90,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:12.795 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.20,"kelembapan":92.90,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:16.794 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":89.60,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:20.827 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":86.30,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:24.827 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":81.90,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:28.822 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":80.90,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:32.836 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.40,"kelembapan":78.60,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:36.822 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":77.50,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:40.791 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.40,"kelembapan":77.40,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:44.818 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":76.80,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:48.825 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":77.00,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:52.814 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.50,"kelembapan":76.00,"status_led":1,"kondisi_suhu":"PANAS"}
08:26:56.828 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.60,"kelembapan":77.40,"status_led":1,"kondisi_suhu":"PANAS"}
08:27:00.791 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.60,"kelembapan":78.20,"status_led":1,"kondisi_suhu":"PANAS"}
08:27:04.802 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.60,"kelembapan":74.70,"status_led":1,"kondisi_suhu":"PANAS"}
```

2. Saat sensor DHT dipanaskan hingga melebihi ambang batas 31.0°C, NodeMCU memublikasikan data peringatan ke topik alert serta otomatis menyalakan LED indikator bahaya.

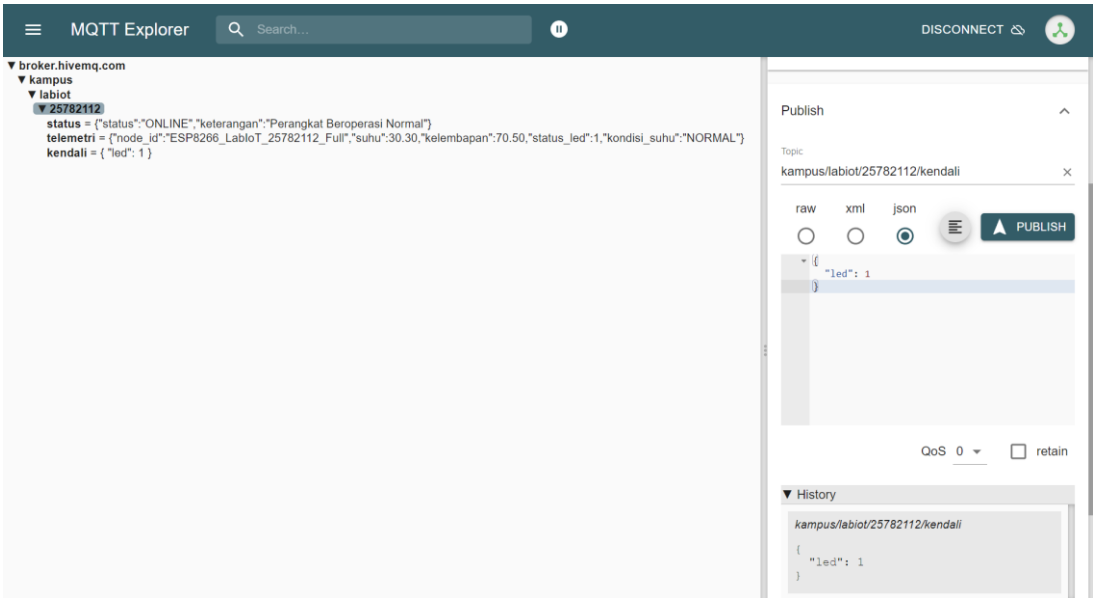


Serial monitor

```
08:28:48.827 -> [ALERT TERKIRIM] {"status":"BAHAYA","parameter":"OVERHEAT","nilai_suhu":31.10}
08:28:48.827 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.10,"kelembapan":79.70,"status_led":1,"kondisi_suhu":"PANAS"}
08:28:52.799 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":32.70,"kelembapan":100.00,"status_led":1,"kondisi_suhu":"PANAS"}
08:28:56.833 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":32.20,"kelembapan":100.00,"status_led":1,"kondisi_suhu":"PANAS"}
08:29:00.842 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.80,"kelembapan":95.50,"status_led":1,"kondisi_suhu":"PANAS"}
08:29:04.806 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.60,"kelembapan":92.60,"status_led":1,"kondisi_suhu":"PANAS"}
08:29:08.834 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.40,"kelembapan":90.20,"status_led":1,"kondisi_suhu":"PANAS"}
08:29:12.797 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":86.60,"status_led":1,"kondisi_suhu":"PANAS"}
08:29:16.824 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":82.20,"status_led":1,"kondisi_suhu":"PANAS"}
08:29:20.828 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":79.40,"status_led":1,"kondisi_suhu":"PANAS"}
08:29:24.816 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":77.90,"status_led":1,"kondisi_suhu":"PANAS"}
08:29:28.801 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.30,"kelembapan":76.90,"status_led":1,"kondisi_suhu":"PANAS"}
```

3. Operator luar tetap dapat mengambil alih kendali manual (*override*) LED melalui topik kendali saat kondisi suhu aman.

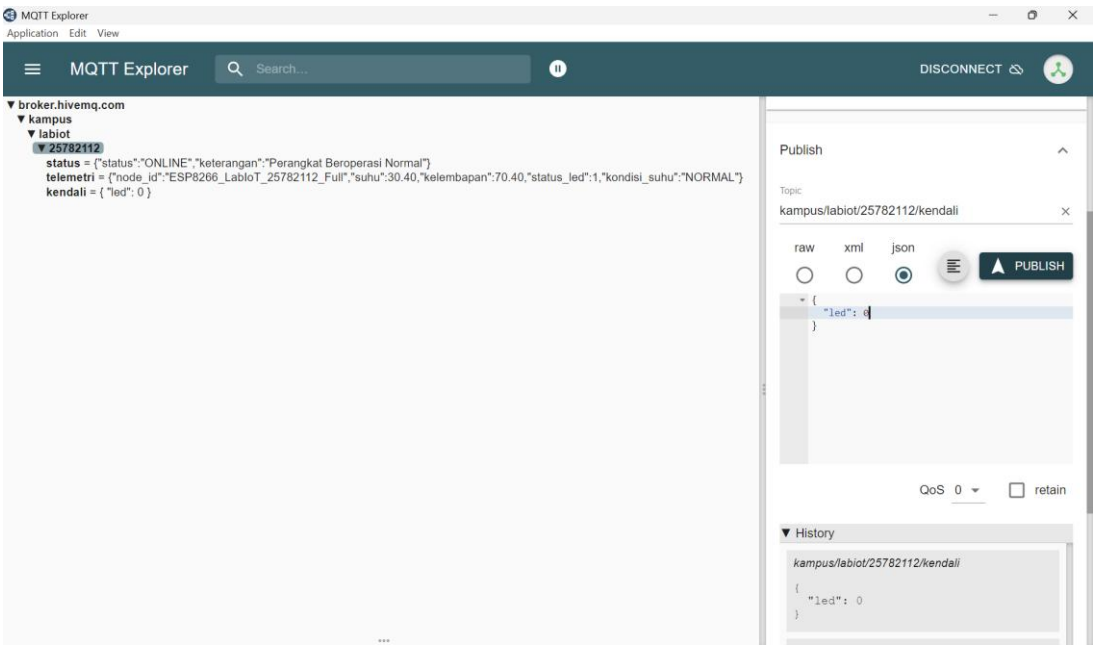
- Menyalakan Led secara manual



Serial monitor

```
08:48:54.247 -> ----------------------------------------
08:48:54.247 -> [KONTROL DITERIMA] Topik: kampus/labiot/25782112/kendali | Payload: {
08:48:54.295 ->   "led": 1
08:48:54.295 -> }
08:48:54.295 -> --> [INDIKATOR] Status LED diubah: MENYALA
08:48:54.295 -> ----------------------------------------
08:48:56.103 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.30,"kelembapan":70.50,"status_led":1,"kondisi_suhu":"NORMAL"}
08:49:00.064 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.40,"kelembapan":70.60,"status_led":1,"kondisi_suhu":"NORMAL"}
08:49:04.109 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.30,"kelembapan":70.50,"status_led":1,"kondisi_suhu":"NORMAL"}
08:49:08.092 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.30,"kelembapan":70.50,"status_led":1,"kondisi_suhu":"NORMAL"}
08:49:12.079 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.30,"kelembapan":70.50,"status_led":1,"kondisi_suhu":"NORMAL"}
```

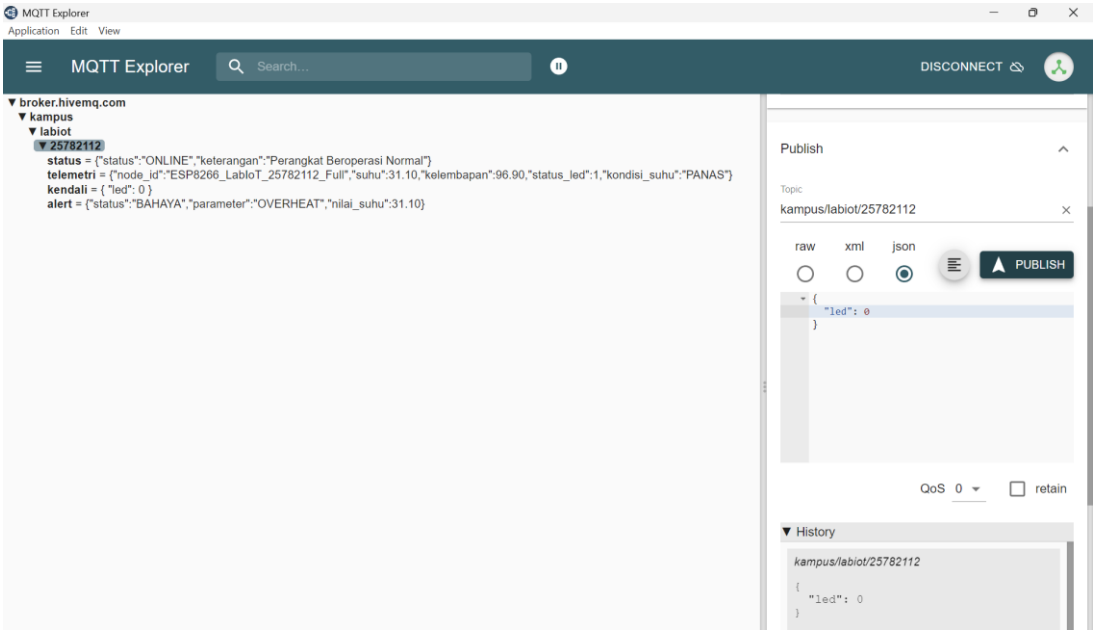
- Mematikan led secara manual



Serial monitor

```
08:51:29.394 -> ----------------------------------------
08:51:29.394 -> [KONTROL DITERIMA] Topik: kampus/labiot/25782112/kendali | Payload: {
08:51:29.442 ->   "led": 0
08:51:29.442 -> }
08:51:29.442 -> --> [INDIKATOR] Status LED diubah: PADAM
08:51:29.442 -> ----------------------------------------
08:51:32.076 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.40,"kelembapan":70.40,"status_led":0,"kondisi_suhu":"NORMAL"}
08:51:36.100 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.40,"kelembapan":70.40,"status_led":0,"kondisi_suhu":"NORMAL"}
08:51:40.078 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.40,"kelembapan":70.30,"status_led":0,"kondisi_suhu":"NORMAL"}
08:51:44.074 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.40,"kelembapan":70.30,"status_led":0,"kondisi_suhu":"NORMAL"}
08:51:48.068 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.40,"kelembapan":70.40,"status_led":0,"kondisi_suhu":"NORMAL"}
```

- Jika keadaan sedang overheat maka kendali manual ditolak



Serial monitor

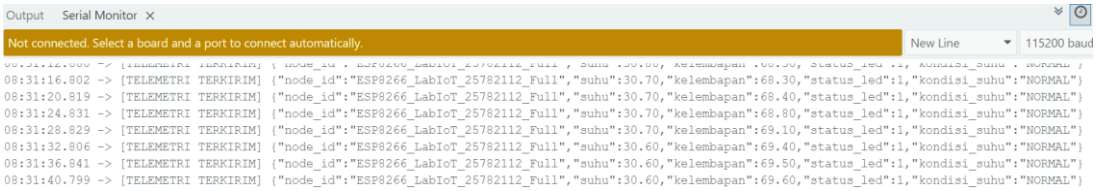
```
08:52:38.036 -> [KONTROL DITERIMA] Topik: kampus/labiot/25782112/kendali | Payload: {
08:52:38.036 ->   "led": 0
08:52:38.036 -> }
08:52:38.036 -> --> [PERINGATAN] Kendali manual ditolak karena kondisi OVERHEAT!
08:52:38.036 -> ----------------------------------------
08:52:40.101 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.20,"kelembapan":98.00,"status_led":1,"kondisi_suhu":"PANAS"}
08:52:44.095 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":31.00,"kelembapan":93.40,"status_led":1,"kondisi_suhu":"NORMAL"}
08:52:48.077 -> [TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782112_Full","suhu":30.90,"kelembapan":90.40,"status_led":1,"kondisi_suhu":"NORMAL"}
```

4. Mekanisme LWT terbukti bekerja saat kabel daya mikrokontroler dicabut mendadak.

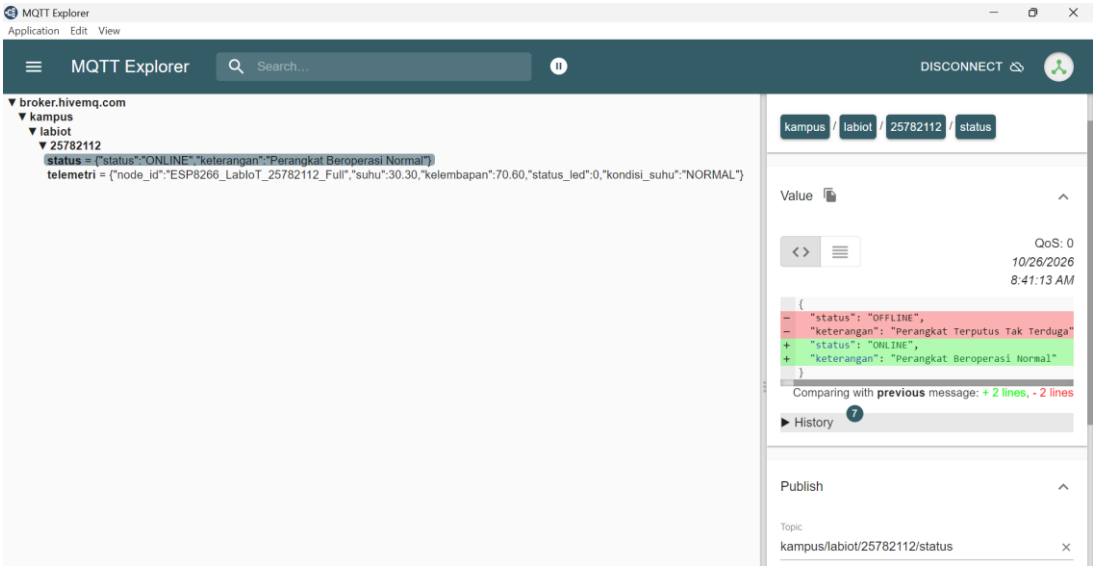
- Jika kabel data dicabut, perhatikan topik kampus/labiot/25782112/status:



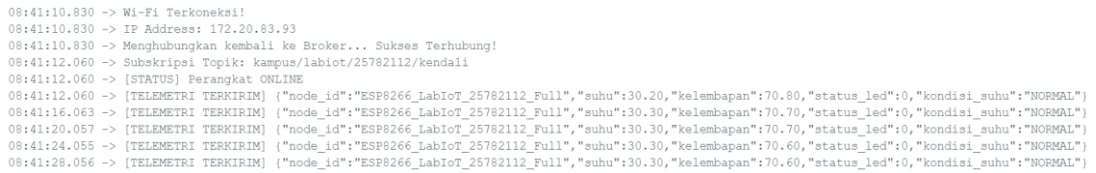
Serial monitor



- Jika kabel data usb terpasang kembali



Serial monitor



5. Seluruh baris kode program ditulis dengan rapi, bebas dari pemanggilan delay() yang memblokir proses, dan memuat komentar dokumentasi teknis yang jelas.

